SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

Creation Date 23-Jun-2008 **Revision Date** 13-Oct-2023 **Revision Number** 6

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: **EZ-RUN Protein Markers and Ladders**
Cat No. : **BP3603-1; BP3603-500**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company

EU entity/business name
Thermo Fisher Scientific
Janssen Pharmaceuticalaan 3a, 2440 Geel,
Belgium

UK entity/business name
Fisher Scientific UK
Bishop Meadow Road,
Loughborough, Leicestershire LE11 5RG,
United Kingdom

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards
Based on available data, the classification criteria are not met

Environmental hazards
Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements

Hazard Statements
EUH210 - Safety data sheet available on request

Precautionary Statements

2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
1,2,3-Propanetriol	56-81-5	200-289-5	< 51	-
Sodium lauryl sulfate	151-21-3	205-788-1	2.0	Flam. Sol. 2 (H228) Acute Tox. 4 (H302) Skin Irrit. 2 (H315) Eye Dam. 1 (H318) Aq. Chronic 3 (H412)
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-	77-86-1	201-064-4	< 1.0	-
2,3-Butanediol, 1,4-dimercapto-, (R*,R*)-	3483-12-3	EEC No. 222-468-7	< 0.76	Acute Tox. 4 (H302) Skin Irrit. 2 (H315) Eye Dam. 1 (H318)
Sodium chloride	7647-14-5	231-598-3	< 0.5	-
Glycine, N,N-1,2-ethanediylbis[N-(carboxymethyl)-, disodium salt, dihydrate	6381-92-6	613-386-6	< 0.05	Acute Tox. 4 (H332) STOT RE 2 (H373)
Sodium azide	26628-22-8	247-852-1	< 0.02	Acute Tox. 2 (H300) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410) (EUH032)
Phenol, 4,4-(3H-1,2-benzoxathiol-3-ylidene)bis[2,6-	62625-28-9	EEC No. 263-653-2	< 0.05	Skin Irrit. 2 (H315) Eye Irrit. 2 (H319)

dibromo-, S,S-dioxide, monosodium salt				STOT SE 3 (H335)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Sodium azide	-	1	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice If symptoms persist, call a physician.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

Ingestion Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

Inhalation Remove to fresh air. Get medical attention immediately if symptoms occur.

Self-Protection of the First Aider No special precautions required.

4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive and/or toxic fumes. Keep product and empty container away from heat and sources of ignition.

Hazardous Combustion Products
Thermal decomposition can lead to release of irritating gases and vapors.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation.

6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid contact with skin, eyes or clothing.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Store in freezer. Keep containers tightly closed in a dry, cool and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany)	Storage Class/LGK 12
Switzerland - Storage of hazardous substances	Storage class - SC 10/12 https://www.kvu.ch/de/themen/stoffe-und-produkte https://www.kvu.ch/fr/themes/substances-et-produits https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work

(Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
1,2,3-Propanetriol		TWA: 10 mg/m³ 8 hr (mist only)	TWA / VME: 10 mg/m³ (8 heures).	TWA: 10 mg/m³ 8 uren	TWA / VLA-ED: 10 mg/m³ (8 horas)
Sodium azide	Skin TWA 0.1 mg/m³ STEL 0.3 mg/m³	Skin TWA 0.1 mg/m³ STEL 0.3 mg/m³	TWA / VME: 0.1 mg/m³ (8 heures). restrictive limit STEL / VLCT: 0.3 mg/m³. restrictive limit Peau	Skin TWA 0.1 mg/m³ STEL 0.3 mg/m³	STEL / VLA-EC: 0.3 mg/m³ (15 minutos). TWA / VLA-ED: 0.1 mg/m³ (8 horas) Piel

Component	Italy	Germany	Portugal	The Netherlands	Finland
1,2,3-Propanetriol		TWA: 200 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 200 mg/m³ (8 Stunden). MAK Höhepunkt: 400 mg/m³	TWA: 10 mg/m³ 8 horas		TWA: 20 mg/m³ 8 tunteina
Sodium azide	TWA: 0.1 mg/m³ 8 ore. Time Weighted Average STEL: 0.3 mg/m³ 15 minuti. Short-term Pelle	MAK 0.2 mg/m³ (inhalable)	STEL: 0.3 mg/m³ 15 minutos Ceiling: 0.29 mg/m³ Ceiling: 0.11 ppm TWA: 0.1 mg/m³ 8 horas Pele	huid STEL: 0.3 mg/m³ 15 minuten TWA: 0.1 mg/m³ 8 uren	TWA: 0.1 mg/m³ 8 tunteina STEL: 0.3 mg/m³ 15 minuutteina Iho

Component	Austria	Denmark	Switzerland	Poland	Norway
1,2,3-Propanetriol			STEL: 100 mg/m³ 15 Minuten TWA: 50 mg/m³ 8 Stunden	TWA: 10 mg/m³ 8 godzinach	
Sodium azide	Haut MAK-KZGW: 0.3 mg/m³ 15 Minuten MAK-TMW: 0.1 mg/m³ 8 Stunden	TWA: 0.1 mg/m³ 8 timer STEL: 0.3 mg/m³ 15 minutter Hud	STEL: 0.4 mg/m³ 15 Minuten TWA: 0.2 mg/m³ 8 Stunden	STEL: 0.3 mg/m³ 15 minutach TWA: 0.1 mg/m³ 8 godzinach	TWA: 0.1 mg/m³ 8 timer STEL: 0.3 mg/m³ 15 minutter. value from the regulation

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
1,2,3-Propanetriol		TWA-GVI: 10 mg/m³ 8 satima.	TWA: 10 mg/m³ 8 hr. (mist)		TWA: 10 mg/m³ 8 hodinách. Ceiling: 15 mg/m³
Sodium azide	TWA: 0.1 mg/m³ STEL : 0.3 mg/m³ Skin notation	kože TWA-GVI: 0.1 mg/m³ 8 satima. STEL-KGVI: 0.3 mg/m³ 15 minutama.	TWA: 0.1 mg/m³ 8 hr. STEL: 0.3 mg/m³ 15 min Skin	Skin-potential for cutaneous absorption STEL: 0.3 mg/m³ TWA: 0.1 mg/m³	TWA: 0.1 mg/m³ 8 hodinách. Potential for cutaneous absorption Ceiling: 0.3 mg/m³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
1,2,3-Propanetriol	TWA: 10 mg/m³ 8 tundides.		TWA: 10 mg/m³		
Sodium azide	Nahk TWA: 0.1 mg/m³ 8 tundides. STEL: 0.3 mg/m³ 15 minutites.	Skin notation TWA: 0.1 mg/m³ 8 hr STEL: 0.3 mg/m³ 15 min	STEL: 0.1 ppm STEL: 0.3 mg/m³ TWA: 0.1 ppm TWA: 0.3 mg/m³	STEL: 0.3 mg/m³ 15 percekben. CK TWA: 0.1 mg/m³ 8 órában. AK	STEL: 0.3 mg/m³ TWA: 0.1 mg/m³ 8 klukkustundum. Skin notation

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Sodium chloride	TWA: 5 mg/m³	TWA: 5 mg/m³ IPRD			
Sodium azide	skin - potential for cutaneous exposure STEL: 0.3 mg/m³ TWA: 0.1 mg/m³	TWA: 0.1 mg/m³ IPRD Oda STEL: 0.3 mg/m³	Possibility of significant uptake through the skin TWA: 0.1 mg/m³ 8 Stunden STEL: 0.3 mg/m³ 15 Minuten	possibility of significant uptake through the skin TWA: 0.1 mg/m³ STEL: 0.3 mg/m³ 15 minuti	Skin notation TWA: 0.1 mg/m³ 8 ore STEL: 0.3 mg/m³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
1,2,3-Propanetriol		TWA: 11 mg/m³	TWA: 200 mg/m³ 8 urah inhalable fraction STEL: 400 mg/m³ 15 minutah inhalable fraction		
Sodium chloride	MAC: 5 mg/m³				
Sodium azide		Ceiling: 0.3 mg/m³ Potential for cutaneous absorption TWA: 0.1 mg/m³	TWA: 0.1 mg/m³ 8 urah Koža STEL: 0.3 mg/m³ 15 minutah	Binding STEL: 0.3 mg/m³ 15 minuter TLV: 0.1 mg/m³ 8 timmar. NGV	Deri TWA: 0.1 mg/m³ 8 saat STEL: 0.3 mg/m³ 15 dakika

Biological limit values
This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)
See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Sodium lauryl sulfate 151-21-3 (2.0)				DNEL = 4060mg/kg bw/day
1,3-Propanediol, 2-amino-2-(hydroxymethyl)- 77-86-1 (< 1.0)				DNEL = 166.7mg/kg bw/day
Sodium chloride 7647-14-5 (< 0.5)		DNEL = 295.52mg/kg bw/day		DNEL = 295.52mg/kg bw/day
Sodium azide 26628-22-8 (< 0.02)				DNEL = 46.7µg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
1,2,3-Propanetriol 56-81-5 (< 51)			DNEL = 56mg/m³	
Sodium lauryl sulfate 151-21-3 (2.0)				DNEL = 285mg/m³
1,3-Propanediol, 2-amino-2-(hydroxymethyl)- 77-86-1 (< 1.0)				DNEL = 117.5mg/m³
Sodium chloride 7647-14-5 (< 0.5)		DNEL = 2068.62mg/m³		DNEL = 2068.62mg/m³
Sodium azide 26628-22-8 (< 0.02)				DNEL = 0.164mg/m³

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)

1,2,3-Propanetriol 56-81-5 (< 51)	PNEC = 0.885mg/L	PNEC = 3.3mg/kg sediment dw	PNEC = 8.85mg/L	PNEC = 1000mg/L	PNEC = 0.141mg/kg soil dw
Sodium lauryl sulfate 151-21-3 (2.0)	PNEC = 0.176mg/L	PNEC = 6.97mg/kg sediment dw	PNEC = 0.055mg/L	PNEC = 1.35mg/L	PNEC = 1.29mg/kg soil dw
1,3-Propanediol, 2-amino-2-(hydroxymethyl)- 77-86-1 (< 1.0)				PNEC = 300mg/L	
Sodium chloride 7647-14-5 (< 0.5)	PNEC = 5mg/L			PNEC = 500mg/L	PNEC = 4.86mg/kg soil dw
Sodium azide 26628-22-8 (< 0.02)	PNEC = 0.35µg/L	PNEC = 16.7µg/kg sediment dw	PNEC = 3.5µg/L	PNEC = 30µg/L	

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
1,2,3-Propanetriol 56-81-5 (< 51)	PNEC = 0.0885mg/L	PNEC = 0.33mg/kg sediment dw			
Sodium lauryl sulfate 151-21-3 (2.0)	PNEC = 0.0176mg/L	PNEC = 0.697mg/kg sediment dw			
Sodium azide 26628-22-8 (< 0.02)	PNEC = 15ng/L	PNEC = 0.72µg/kg sediment dw	PNEC = 150ng/L		

8.2. Exposure controls

Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Personal protective equipment

Eye Protection Wear safety glasses with side shields (or goggles) (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Natural rubber Nitrile rubber Neoprene PVC	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Particle filter

Small scale/Laboratory use Maintain adequate ventilation

Environmental exposure controls No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Blue	
Odor	Odorless	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	No information available	**Method -** No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	7.5 - 8.0	
Viscosity	No data available	
Water Solubility	Miscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
1,2,3-Propanetriol	-1.75	
Sodium lauryl sulfate	1.6	
Vapor Pressure	No data available	
Density / Specific Gravity	No data available	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under recommended storage conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization	No information available.
Hazardous Reactions	None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat.

10.5. Incompatible materials

Strong oxidizing agents.

10.6. Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral Based on available data, the classification criteria are not met
Dermal Based on available data, the classification criteria are not met
Inhalation Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
1,2,3-Propanetriol	12600 mg/kg (Rat)	> 10 g/kg (Rabbit)	> 2.75 mg/L/4h (Rat)(mist)
Sodium lauryl sulfate	LD50 = 1288 mg/kg (Rat)	LD50 = 200 mg/kg (Rabbit)	LC50 > 3900 mg/m³ (Rat) 1 h
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-	LD50 = 5900 mg/kg (Rat)	LD50 > 5000 mg/kg (Rat)	-
2,3-Butanediol, 1,4-dimercapto-, (R*,R*)-	400 mg/kg (Rat)	-	-
Sodium chloride	LD50 = 3 g/kg (Rat)	LD50 > 10000 mg/kg (Rabbit)	LC50 > 42 mg/L (Rat) 1 h
Sodium azide	LD50 = 27 mg/kg (Rat)	-	LC50 0.054 - 0.52 mg/L (Rat) 4 h

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects,both acute and delayed No information available.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects Do not empty into drains. .

Component	Freshwater Fish	Water Flea	Freshwater Algae
1,2,3-Propanetriol	LC50: 51 - 57 mL/L, 96h static (Oncorhynchus mykiss)		
Sodium lauryl sulfate	1.31 mg/L LC50 96 h 9.9-20.1 mg/L LC50 96 h 4.5 mg/L LC50 96 h 4.62 mg/L LC50 96 h 7.97 mg/L LC50 96 h 10.2-22.5 mg/L LC50 96 h 10.8-16.6 mg/L LC50 96 h 13.5-18.3 mg/L LC50 96 h 15-18.9 mg/L LC50 96 h 22.1-22.8 mg/L LC50 96 h 4.06-5.75 mg/L LC50 96 h 4.2-4.8 mg/L LC50 96 h 4.3-8.5 mg/L LC50 96 h 5.8-7.5 mg/L LC50 96 h 6.2-9.6 mg/L LC50 96 h 8-12.5 mg/L LC50 96 h 4.2 mg/L LC50 96 h	EC50: = 1.8 mg/L, 48h (Daphnia magna)	EC50: 3.59 - 15.6 mg/L, 96h static (Pseudokirchneriella subcapitata) EC50: = 117 mg/L, 96h (Pseudokirchneriella subcapitata) EC50: 30 - 100 mg/L, 96h (Desmodesmus subspicatus) EC50: = 53 mg/L, 72h (Desmodesmus subspicatus)
Sodium chloride	Pimephals prome: LC50: 7650 mg/L/96h	EC50: 1000 mg/L/48h	
Sodium azide	LC50: = 0.7 mg/L, 96h (Lepomis macrochirus) LC50: = 0.8 mg/L, 96h (Oncorhynchus mykiss) LC50: = 5.46 mg/L, 96h flow-through (Pimephales promelas)		

Component	Microtox	M-Factor
Sodium lauryl sulfate	= 0.46 mg/L EC50 Photobacterium phosphoreum 30 min = 0.72 mg/L EC50 Photobacterium phosphoreum 15 min = 1.19 mg/L EC50 Photobacterium phosphoreum 5 min	
Sodium azide		1

12.2. Persistence and degradability

Persistence Miscible with water, Persistence is unlikely, based on information available.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)

1,2,3-Propanetriol	-1.75	No data available
Sodium lauryl sulfate	1.6	No data available

12.4. Mobility in soil The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

12.5. Results of PBT and vPvB assessment No data available for assessment.

12.6. Endocrine disrupting properties
Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects
Persistent Organic Pollutant This product does not contain any known or suspected substance
Ozone Depletion Potential This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC) According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

Switzerland - Waste Ordinance Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

ADR Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

IATA Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories
Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
1,2,3-Propanetriol	56-81-5	200-289-5	-	-	X	X	KE-29297	X	X
Sodium lauryl sulfate	151-21-3	205-788-1	-	-	X	X	KE-21884	X	X
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-	77-86-1	201-064-4	-	-	X	X	KE-01403	X	X
2,3-Butanediol, 1,4-dimercapto-, (R*,R*)-	3483-12-3	222-468-7	-	-	X	X	-	-	-
Sodium chloride	7647-14-5	231-598-3	-	-	X	X	KE-31387	X	X
Glycine, N,N-1,2-ethanediylbis[N-(carboxy methyl)-, disodium salt, dihydrate	6381-92-6	-	-	-	X	X	-	-	-
Sodium azide	26628-22-8	247-852-1	-	-	X	X	KE-31357	X	X
Phenol, 4,4-(3H-1,2-benzoxathiol-3-ylidene)bis[2,6-dibromo-, S,S-dioxide, monosodium salt	62625-28-9	263-653-2	-	-	X	X	-	-	-

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
1,2,3-Propanetriol	56-81-5	X	ACTIVE	X	-	X	X	X
Sodium lauryl sulfate	151-21-3	X	ACTIVE	X	-	X	X	X
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-	77-86-1	X	ACTIVE	X	-	X	X	X
2,3-Butanediol, 1,4-dimercapto-, (R*,R*)-	3483-12-3	X	ACTIVE	X	-	X	X	X
Sodium chloride	7647-14-5	X	ACTIVE	X	-	X	X	X
Glycine, N,N-1,2-ethanediylbis[N-(carboxy methyl)-, disodium salt, dihydrate	6381-92-6	-	-	X	-	X	X	X
Sodium azide	26628-22-8	X	ACTIVE	X	-	X	X	X
Phenol, 4,4-(3H-1,2-benzoxathiol-3-ylidene)bis[2,6-dibromo-, S,S-dioxide, monosodium salt	62625-28-9	X	ACTIVE	X	-	-	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH Not applicable

Component	CAS No	REACH (1907/2006) -	REACH (1907/2006) -	REACH Regulation (EC

		Annex XIV - Substances Subject to Authorization	Annex XVII - Restrictions on Certain Dangerous Substances	1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
1,2,3-Propanetriol	56-81-5	-	-	-
Sodium lauryl sulfate	151-21-3	-	-	-
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-	77-86-1	-	-	-
2,3-Butanediol, 1,4-dimercapto-, (R*,R*)-	3483-12-3	-	-	-
Sodium chloride	7647-14-5	-	-	-
Glycine, N,N-1,2-ethanediylbis[N-(carboxymethyl)-, disodium salt, dihydrate	6381-92-6	-	-	-
Sodium azide	26628-22-8	-	-	-
Phenol, 4,4-(3H-1,2-benzoxathiol-3-ylidene)bis[2,6-dibromo-, S,S-dioxide, monosodium salt	62625-28-9	-	-	-

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
1,2,3-Propanetriol	56-81-5	Not applicable	Not applicable
Sodium lauryl sulfate	151-21-3	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-	77-86-1	Not applicable	Not applicable
2,3-Butanediol, 1,4-dimercapto-, (R*,R*)-	3483-12-3	Not applicable	Not applicable
Sodium chloride	7647-14-5	Not applicable	Not applicable
Glycine, N,N-1,2-ethanediylbis[N-(carboxymethyl)-, disodium salt, dihydrate	6381-92-6	Not applicable	Not applicable
Sodium azide	26628-22-8	Not applicable	Not applicable
Phenol, 4,4-(3H-1,2-benzoxathiol-3-ylidene)bis[2,6-dibromo-, S,S-dioxide, monosodium salt	62625-28-9	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals
Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 1 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class

1,2,3-Propanetriol	WGK1	
Sodium lauryl sulfate	WGK2	
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-	WGK1	
2,3-Butanediol, 1,4-dimercapto-, (R*,R*)-	WGK2	
Sodium chloride	WGK1	
Glycine, N,N-1,2-ethanediylbis[N-(carboxymethyl)-, disodium salt, dihydrate	WGK2	
Sodium azide	WGK2	

Component	France - INRS (Tables of occupational diseases)
Sodium chloride	Tableaux des maladies professionnelles (TMP) - RG 78

Swiss Regulations
Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Sodium lauryl sulfate 151-21-3 (2.0)	Prohibited and Restricted Substances		
Sodium chloride 7647-14-5 (< 0.5)	Prohibited and Restricted Substances		
Glycine, N,N-1,2-ethanediylbis[N-(carboxymethyl)-, disodium salt, dihydrate 6381-92-6 (< 0.05)	Prohibited and Restricted Substances		
Phenol, 4,4-(3H-1,2-benzoxathiol-3-ylidene)bis[2,6-dibromo-, S,S-dioxide, monosodium salt 62625-28-9 (< 0.05)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3
H228 - Flammable solid
H300 - Fatal if swallowed
H302 - Harmful if swallowed
H311 - Toxic in contact with skin
H315 - Causes skin irritation
H319 - Causes serious eye irritation
H332 - Harmful if inhaled
H335 - May cause respiratory irritation
H400 - Very toxic to aquatic life
H410 - Very toxic to aquatic life with long lasting effects
EUH032 - Contact with acids liberates very toxic gas

Legend

CAS - Chemical Abstracts Service

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical

Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Creation Date	23-Jun-2008
Revision Date	13-Oct-2023
Revision Summary	Not applicable.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer
The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet